Exercise 2: Moving Shapes 

The idea of the exercise is that we should try to get shapes moving either by themselves, by using lopped in **`draw()`** and re-positioning the shape's position, or through interaction with our mouse/trackpad, by using **`mouseX`** and **`mouseY`** .

Create a new sketch for each part of the exercise.

Part-exercise 1: Falling ball on repeat

★ Above the **`setup()`** function, we create a global variable that holds our ball's y-axis value. We can call it **`circleY`** and give it the value 0. We also need to give our ball a velocity. Let's call this variable **`speed`** and give it an initial value of 1.

★ Start by creating a new canvas **`createCanvas()`** in **`setup()`**.
You decide the height and width yourself.
★ Make your background black with **`background()`**. Type this and subsequent code in **`draw()`**.

★ Now make a ball using **`circle()`**.

This function has three parameters **`circle(x, y, d)`**. Set the x value to a value less than the height and width of your canvas, y should be our global variable **`circleY`**, and d, which specifies the diameter of our ball, should also just be a value less than the height and width of your canvas. Give it a try or see example.

Example

```
// If your canvas is 200 x 200 pixels

 circle(200, circleY, 50);
```

★ Now we want our ball to fall from top to bottom. We do this by changing its value, specifically its y-axis value, which is the value our variable `circleY` holds.
We want this value to increase, so make a statement that lets
`circleY` increase by 1.

If we run our code now, we'll see our ball drop down and out of our canvas.

★ As a final thing, we want as soon as our ball hits the bottom of our canvas, it drops again from the top of our canvas.

To do this we need a 'check'.
We need to check that if our ball's y-axis value is greater than the height of our canvas, then we want to reset our ball's y-axis value.

★ Do you have an idea for how we do this? Just read it again and give it a shot.
Otherwise see below:

```
if (circleY > height) {
  circleY = 0;
}
```

Try writing comments to your code. Do you understand what you just did and what is happening? Otherwise, ask!